

A MIXED RADIAL MOORE GRAPH WITH PARAMETERS (5, 2, 2)

ABSTRACT. We give an explicit totally (5, 2)-regular mixed graph on $52 = M(5, 2, 2)$ vertices with radius 2 and diameter 3. Hence an $\text{RM}(5, 2, 2)$ mixed radial Moore graph exists, settling a case marked open in Table 3 of Ceresuela, López, and Chemisana and still marked open in Table 4.2 of Abiad, Ceresuela, and López; among the many parameter pairs left unresolved in the latter table, $(r, z) = (5, 2)$ is the one of least Moore bound, $M(5, 2, 2) = 52$. The graph has unique central vertex 0, total status 5741, and status 1-norm $N_1 = 801$. No optimality assertion for N_1 is made.

1. THE CONSTRUCTION

A mixed graph has undirected edges and directed arcs. Distances are source-to-target distances: edges may be traversed in either direction, whereas arcs may be traversed only forward. The center $C(G)$ is the set of vertices of minimum out-eccentricity. A mixed graph is totally (r, z) -regular if every vertex has undirected degree r , indegree z , and outdegree z . Following [1], an $\text{RM}(r, z, k)$ graph is a totally (r, z) -regular mixed graph of radius k , diameter $k + 1$, and order equal to the mixed Moore bound $M(r, z, k)$.

For $k = 2$, the Moore-tree count is

$$(1) \quad M(r, z, 2) = 1 + (r + z) + r(r - 1 + z) + z(r + z) = (r + z)^2 + z + 1.$$

Thus $M(5, 2, 2) = 52$. The existence question for this parameter pair is marked unresolved in Table 3 of [1] and Table 4.2 of [2]. For $k = 2$, Ceresuela, López, and Chemisana constructed $\text{RM}(r, 1, 2)$ with $r \geq 1$ and $\text{RM}(1, z, 2)$ with $z \geq 1$ [1], and Abiad, Ceresuela, and López settled the six further cases $(r, z) = (2, 2), (2, 3), (3, 2), (3, 3), (4, 2), (2, 4)$ by integer programming [2]. The result below gives a positive answer to the $(5, 2, 2)$ case of Question 1 in [1].

For a vertex v , let

$$s(v) = \sum_{u \in V(G)} d(v, u)$$

be its status. The hypothetical mixed Moore distance layers for these parameters have sizes 1, 7, 44, so the corresponding vertex status is

$$(2) \quad \sigma_{5,2,2} = 7 + 2 \cdot 44 = 95.$$

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We therefore use

$$N_1(G) = \sum_{v \in V(G)} |s(v) - 95|,$$

in agreement with the status norm of [1].

Theorem 1. *Let G be the mixed graph on $V(G) = \{0, 1, \dots, 51\}$ whose undirected neighborhoods $U(v)$ and outgoing-arc heads $A^+(v)$ are listed in the adjacency certificate below. Then G is an RM(5, 2, 2) graph. Moreover,*

$$C(G) = \{0\}, \quad \sum_{v \in V(G)} s(v) = 5741, \quad N_1(G) = 801.$$

Proof. The certificate has no loops, directed digons, or edge-arc overlaps. Undirected adjacency is symmetric, every row contains five undirected neighbors and two outgoing arcs, and every vertex appears as an arc head exactly twice. Thus G is totally (5, 2)-regular, with 130 edges and 104 arcs.

The vertices at distance one from 0 are $\{1, \dots, 7\}$. After the parent edge is omitted for vertices $1, \dots, 5$, their admissible second-layer descendants, together with those of 6 and 7, are respectively

$$\begin{aligned} &\{8, \dots, 13\}, \{14, \dots, 19\}, \{20, \dots, 25\}, \{26, \dots, 31\}, \{32, \dots, 37\}, \\ &\{38, \dots, 44\}, \{45, \dots, 51\}. \end{aligned}$$

These sets are disjoint and exhaust $V(G) \setminus \{0, \dots, 7\}$. Hence the distance layers from 0 have sizes 1, 7, 44, and $\text{ecc}^+(0) = 2$.

Breadth-first search from all 52 sources in the out-neighborhood relation $U(v) \cup A^+(v)$ gives

$$\#\{(u, v) : d(u, v) = i\} = \begin{cases} 52, & i = 0, \\ 364, & i = 1, \\ 1487, & i = 2, \\ 801, & i = 3, \end{cases}$$

and

$$\text{ecc}^+(0) = 2, \quad \text{ecc}^+(v) = 3 \quad (1 \leq v \leq 51).$$

Consequently G has radius 2, diameter 3, and center $\{0\}$. Together with (1), this proves that $G \in \text{RM}(5, 2, 2)$.

Every vertex has seven targets at distance one. If $q(v)$ is the number of targets at distance three from v , then the remaining $44 - q(v)$ targets are at distance two, and hence

$$s(v) = 7 + 2(44 - q(v)) + 3q(v) = 95 + q(v).$$

Since $\sum_v q(v) = 801$, we obtain

$$\sum_v s(v) = 52 \cdot 95 + 801 = 5741$$

and, by (2),

$$N_1(G) = \sum_v |s(v) - 95| = \sum_v q(v) = 801.$$

□

Verification. The adjacency certificate below is self-contained: the structural conditions, the all-pairs distances, the eccentricities, the statuses, and N_1 are all finite computations on the 52 printed rows, and we performed them by computer directly from that table. No ancillary file is needed to reproduce the checks.

Adjacency certificate. For each vertex v , the following table gives the undirected neighbors $U(v)$ and the heads $A^+(v)$ of its outgoing arcs.

v	$U(v)$	$A^+(v)$	v	$U(v)$	$A^+(v)$
0	1, 2, 3, 4, 5	6, 7	26	4, 15, 22, 30, 41	31, 38
1	0, 8, 9, 10, 11	12, 13	27	4, 15, 24, 36, 40	13, 22
2	0, 14, 15, 16, 17	18, 19	28	4, 19, 20, 39, 51	5, 34
3	0, 20, 21, 22, 23	24, 25	29	4, 22, 32, 33, 51	2, 44
4	0, 26, 27, 28, 29	30, 31	30	14, 21, 26, 39, 51	20, 42
5	0, 32, 33, 34, 35	36, 37	31	10, 32, 37, 43, 44	8, 34
6	38, 39, 40, 41, 42	43, 44	32	5, 11, 29, 31, 47	9, 46
7	45, 46, 47, 48, 49	50, 51	33	5, 9, 18, 22, 29	15, 19
8	1, 36, 43, 48, 50	14, 47	34	5, 13, 16, 19, 39	17, 22
9	1, 25, 33, 35, 43	16, 50	35	5, 9, 14, 46, 50	18, 27
10	1, 21, 23, 31, 44	4, 33	36	8, 23, 25, 27, 49	10, 12
11	1, 12, 15, 32, 40	23, 35	37	31, 38, 42, 45, 46	16, 30
12	11, 14, 16, 44, 46	32, 51	38	6, 24, 37, 43, 49	0, 29
13	16, 25, 34, 46, 50	41, 45	39	6, 28, 30, 34, 51	27, 48
14	2, 12, 30, 35, 45	26, 32	40	6, 11, 24, 27, 50	9, 29
15	2, 11, 23, 26, 27	17, 40	41	6, 17, 22, 26, 48	21, 23
16	2, 12, 13, 34, 44	15, 41	42	6, 18, 37, 47, 49	8, 25
17	2, 19, 20, 41, 47	24, 45	43	8, 9, 25, 31, 38	4, 48
18	19, 21, 33, 42, 48	1, 5	44	10, 12, 16, 31, 45	43, 49
19	17, 18, 28, 34, 50	14, 42	45	7, 14, 37, 44, 48	10, 11
20	3, 17, 21, 28, 47	33, 39	46	7, 12, 13, 35, 37	3, 6
21	3, 10, 18, 20, 30	1, 7	47	7, 17, 20, 32, 42	3, 36
22	3, 26, 29, 33, 41	11, 28	48	7, 8, 18, 41, 45	28, 35
23	3, 10, 15, 24, 36	0, 26	49	7, 25, 36, 38, 42	20, 47
24	23, 27, 38, 40, 51	2, 37	50	8, 13, 19, 35, 40	21, 46
25	9, 13, 36, 43, 49	38, 39	51	24, 28, 29, 30, 39	40, 49

REFERENCES

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